

Lecture Notes: Introduction to Astrophysics and Cosmology

Based on lectures by **Dr. Shmuel Bialy** in 2025–26
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These notes were typed in the 2026 winter semester *Introduction to Astrophysics and Cosmology* course taught by **Dr. Shmuel Bialy** at the Technion.

The lectures were given in Hebrew and were live-translated by me to English. The document is provided as is and likely contains many errors.

If you find any mistakes or typos please let me know at danielbreger@campus.technion.ac.il.

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Lecture 1.

Thu, October 30, 2025

1 introduction

Since astrophysics is a very wide subject we won't be able to dive deep on each subject. The course is introductory so we'll be skipping some things like full solutions to integrals and the such.

1.1 So what is astrophysics?

Astrophysics and Cosmology could be broken down into:

Physics of stars

Solar systems planets, comets...

ISM Inter-Stellar Medium, Inter-Galactic Medium

Cosmology development, the big bang, dark energy/matter

Exotic objects black holes, neutron stars, white dwarfs...

Relativity gravitational waves, gravitational lensing

Galaxies types, development, active galactic nuclei, quasars

Explosions novae, supernovae, gamma ray bursts

and the methods used to research those things include

Observations Telescopes IR, radio, xray, gamma rays...

Gravitational waves

Neutrinos

Other particles (via probes)

Labs

Computer simulations hydrodynamics, N-body simulations...

Analytical theory

We'll focus mostly on the analytical theory, but we'll mention the others too.

Mon, November 3rd, 2025

1.2 Regular (Baryonic) Matter

In our normal day-to-day life we meet protons, neutrons, and electrons. There are other particles we come across, such as neutrinos, muons, taus, etc. Another particle we meet often is the photon, and some others we won't get into.

When the universe came into being in the big bang quarks and photons were created. Those quarks then combined into protons and neutrons. Protons are basically hydrogen nuclei, so hydrogen was created in the big bang. For a short period of time after the big bang there were conditions for nuclear fusion across the early universe, so about 10% of the matter created from the big bang ended up being helium, and in terms of mass it was about 25% (Because Helium is heavier than hydrogen). Elements heavier than helium weren't created at any significant quantities in the big bang. If we plot this on a graph:

The portion of mass of hydrogen of the total mass is denoted by X , hydrogen by Y , and metals, which are all the elements heavier than helium and are denoted by $Z = \frac{M_{\text{metals}}}{M}$. For our sun:

$$X_{\odot} = 0.7381 \quad (2.1)$$

$$Y_{\odot} = 0.2485 \quad (2.2)$$

$$Z_{\odot} = 0.0134 \quad (2.3)$$

The above was a discussion for gas and plasma. There is another kind of matter in space called dust. Dust is similar to the dust we know. It's particles about a micrometer in diameter made of water, carbon, silica, etc. Why do we mention this? In terms of the mass of the metals in the universe, about half of it is dust. Moreover, dust is a significant factor in observations of the universe, either in interfering in our ability to see things or in creating conditions that allow us to infer information. Dust also aids the creation of molecules by speeding up processes which are very slow in gasses.

An important definition for us will be the 'average mass' which we'll use in many formulas. Assume we have a box filled with hydrogen and some helium and some metals, all with the relative quantities that are in the sun. What would the average mass per particle be? Approximately a proton mass. And if the gas in the box is ionized? 0.5 proton masses. This is because we doubled the number of particles by freeing the electrons, which have approximately 0 mass relatively to the protons. This was a back of the napkin approximation just for sanity check and to have an idea of what we're expecting. The exact calculations:

$$\bar{m} = \frac{\sum_j N_j m_j}{\sum_j N_j} \quad (2.4)$$

Where j is each element type, N is their number, and m their mass. Rewriting in terms of X, Y, Z :

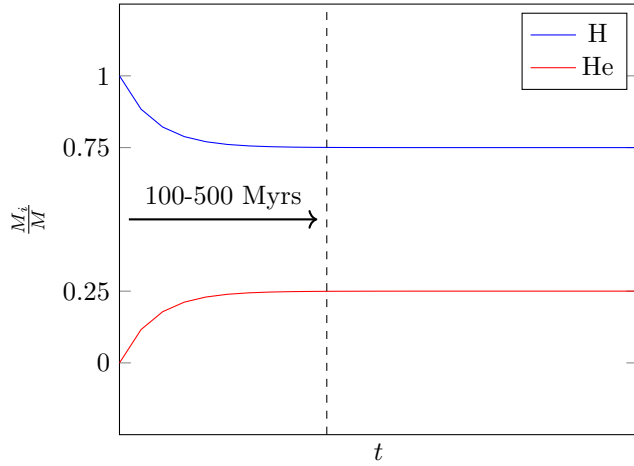


Figure 2.1: The portion of mass of the universe by element in the universe's early times.

$$\frac{1}{\bar{m}} = \frac{\sum N_j}{\sum N_j m_j} \quad (2.5)$$

$$= \frac{M_j}{M_{tot}} \cdot \sum \frac{N_j}{M_j} \quad (2.6)$$

$$= \frac{1}{m_H} \left(X + \frac{1}{4}Y + \frac{1}{A_Z}Z \right) \quad (2.7)$$

where

$$M_j = \begin{cases} N_H m_H & H \\ N_{He} \cdot 4m_H & He \\ N_Z A_Z & else \end{cases} \quad (2.8)$$

Sometimes the following definition is used:

$$\mu = \frac{\bar{m}}{m_H} \quad (2.9)$$

and if we plug in the numbers for the sun we indeed find that it's very close to 1. We won't do this explicitly because there's a correction for the formula for a non-neutral gas, which our sun is not. For a completely ionized gas, i.e all the electrons are free, we just need to change N_j :

$$N_j = \begin{cases} 1 \rightarrow 2 & H \\ 1 \rightarrow 3 & He \\ 1 \rightarrow \langle \frac{A_Z}{2} \rangle & Else \end{cases} \quad (2.10)$$

Which gives us, for a completely ionized gas:

$$\mu = \left[2X + \frac{3}{4}Y + \frac{1}{2}Z \right]^{-1} \quad (2.11)$$

and if you insert the numbers for our sun we find

$$\langle A \rangle = 15 - 16 \quad (2.12)$$

$$\mu_{\odot} \simeq 0.6 \quad (2.13)$$

Thu, November 6th, 2025

1.3 The meaning of the symbol $\simeq$

The symbol $\simeq$ usually implies a good understanding of the physics behind the calculations. It says there's a good understanding of what can be considered negligible and what can't. The luminosity of the sun follows

$$L_{\odot} = \dot{E}_{\text{nuc}} + |\dot{E}_G| \quad (3.1)$$

Observationally we can measure the luminosity and find

$$L_{\odot} = 3.8 \times 10^{33} \text{ erg s}^{-1} \quad (3.2)$$

and we can say that

$$L_{\odot} \simeq \dot{E}_{\text{nuc}} \quad (3.3)$$

Let's look at the total energy the Sun will output in its entire life E_{tot} .

$$E_{\text{tot}} = L_{\odot} \cdot t_{\odot} \quad (3.4)$$

$$\simeq (3.8 \times 10^{33} \text{ erg s}^{-1})(4.6 \text{ Gyr}) \quad (3.5)$$

$$= 5.6 \times 10^{50} \text{ erg} \quad (3.6)$$

Let's look at what contributes more to the energy of the sun, the gravitational or the nuclear energy.

$$|E_G| = \int \frac{GM(r)}{r} dm \quad (3.7)$$

$$= \int 4\pi\rho GM(r)r dr \quad (3.8)$$

$$(\text{assuming } \rho = \text{const.}) = \frac{3}{5}G \frac{M_{\odot}^2}{R_{\odot}} \quad (3.9)$$

$$= 2 \times 10^{48} \text{ erg} \quad (3.10)$$

To estimate the nuclear energy the sun produces we can look at the nuclear fusion process in which $4H \rightarrow He$ which gives a mass difference of

$$\Delta m = M(4H) - M(He) \quad (3.11)$$

$$\rightarrow \Delta E = \Delta mc^2 \quad (3.12)$$

defining

$$\eta = \frac{\Delta E}{M_{He}c^2} = 0.7\% \quad (3.13)$$

then the nuclear energy produced in the sun is

$$\eta \cdot M_{\odot}c^2 \quad (3.14)$$

and using this we can find the age of the sun

$$t_{\odot} = \frac{E_{\text{nuc}}}{L_{\odot}} \approx 100 \text{ Gyr} \quad (3.15)$$

but since the sun hasn't burned through all the hydrogen it has access to, we can replace

$$M_{\odot} \rightarrow M_{\text{core}} \approx 5\% M_{\odot} \quad (3.16)$$

which would give us a better estimate.

2 The Structure of Stars

We'll use the Sun as an example, then we'll hit equations.

2.1 The Sun

Our goal in the coming weeks will be to derive differential equations to figure out how the different profiles behave. As for some numbers:

$$\begin{aligned} R_{\odot} &= 7 \times 10^{10} \text{ cm} & T_C &= 1.5 \times 10^7 \text{ K} \\ \rho_C &= 150 \text{ g cm}^{-3} & \bar{\rho} &= 1.4 \text{ g cm}^{-3} \\ P_C &= 10^{11} \text{ Atm} & M_{\odot} &= 2 \times 10^{33} \text{ g} \\ L_{\odot} &= 3.8 \times 10^{33} \text{ erg/s} \end{aligned}$$

2.2 Star Structure Equations

The equations describing the structure of stars are:

$$\frac{dP}{dr} = -G \frac{M_r \rho}{r^2} \quad (3.17)$$

$$\frac{dM_r}{dr} = 4\pi r^2 \rho \quad (3.18)$$

$$\frac{dL_r}{dr} = 4\pi r^2 \rho \epsilon \quad (3.19)$$

$$\frac{dT}{dr} = -\frac{3\kappa\rho}{4acT^3} \frac{L_r}{4\pi r^2} \quad (3.20)$$

The first equation is called the hydrostatic equation and is equivalent to conservation of momentum, the second is the continuum equation and is equivalent to conservation of mass, the third is the energy production equation and is equivalent to conservation of energy, and the final one is the radiation diffusion equation.

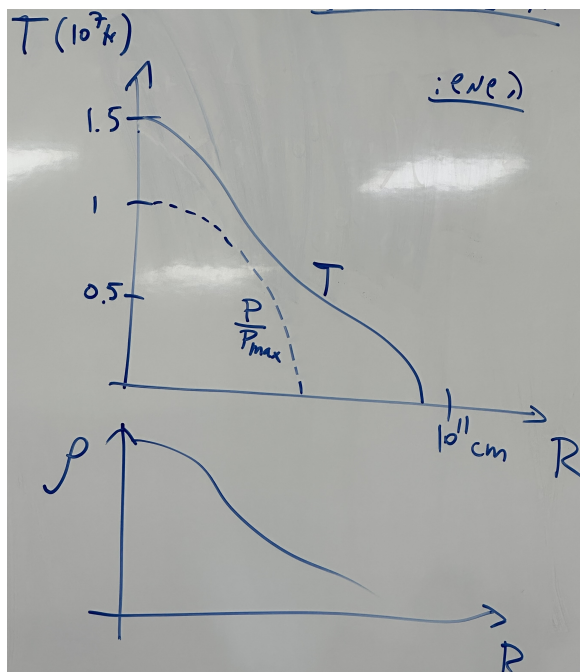


Figure 3.1: The temperature in the sun as a function of the distance from it core, the relative pressure compared to the core, and the density.

2.3 The Hydrostatic Equation

The equation can be derived from an equality of forces. We look at a shell at radius r with mass dm and ask which forces act on it. On one hand there's gravitation pulling it inwards and on the other side there's a force resulting from a difference in pressure.

$$|dF_g| = \frac{GM_r dm}{r^2} \quad (3.21)$$

$$= \frac{GM_r}{r^2} 4\pi r^2 \rho dr \quad (3.22)$$

$$|dF_P| = A \cdot (P(r) - P(r + dr)) \quad (3.23)$$

$$= 4\pi r^2 \left(-\frac{dP}{dr} dr \right) \quad (3.24)$$

equating the two forces we can find

$$\frac{dP}{dr} = -\rho \frac{GM_r}{r^2} \quad (3.25)$$

which is indeed the hydrostatic equation.

Applying The Equation To Earth

As a back of the envelope calculation, we can use the equation to find the gravitational acceleration of earth. Noticing that

$$\frac{GM_r}{r^2} = g \quad (3.26)$$

we can rewrite the hydrostatic equation as

$$\frac{dP}{dr} = -\rho g \quad (3.27)$$

We can also find the speed of sound

$$c_s = \sqrt{\gamma \frac{dP}{d\rho}} \quad (3.28)$$

which means

$$\frac{dP}{d\rho} \approx c_s^2 \quad (3.29)$$

so

$$g \sim \frac{dP}{d\rho} \frac{1}{h} \quad (3.30)$$

$$= \frac{(340 \text{ m s}^{-1})^2}{10 \text{ km}} \quad (3.31)$$

$$= 10 \text{ m/s}^2 \quad (3.32)$$

Mon, November 10th, 2025

2.4 The State Equation

The gas in stars is to a good approximation an ideal gas so we can use the state equation of an ideal gas

$$P = Nk_B T \quad (4.1)$$

we can multiply and divide by the average mass of each particle

$$P = \frac{n\bar{m}}{\bar{m}} k_B T = \frac{\rho}{m_p \mu} k_B T \quad (4.2)$$

A quick aside on why the state equation is true

This is taken from Feynman's lectures on physics volume 2 chapter 39-2. Assume we have a wall of area A and particles with density n , each having velocity v_x and they impact the wall head on. As we know $F = \frac{dp}{dt}$. The distance the particles travel over a period of time of δt is $\delta x = v_x \delta t$. So the rate at which particles hit over the same time period is $\delta N = \frac{1}{2} n A v_x \delta t$. Each particles transfers a momentum of $2p_1 = 2mv$ therefore the total momentum transferred is

$$\delta p = mn A v_x^2 \delta t \quad (4.3)$$

therefore the force is

$$F = \frac{dp}{dt} = mn A v_x^2 \quad (4.4)$$

and the pressure is

$$P = \frac{F}{A} = mn v_x^2 \quad (4.5)$$

and if the velocity isn't a constant but rather there's a distribution of velocities then

$$P = mn \langle v_x^2 \rangle \quad (4.6)$$

when in 3 dimensions

$$\langle v^2 \rangle = \langle v_x^2 \rangle + \langle v_y^2 \rangle + \langle v_z^2 \rangle = 3 \langle v_x^2 \rangle \quad (4.7)$$

So the pressure is

$$P = \frac{1}{3} mn \langle v^2 \rangle \quad (4.8)$$

If we define temperature as follows

$$\frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_B T \quad (4.9)$$

and then insert this into our equation for pressure we get

$$P = nk_B T \quad (4.10)$$

Thermal energy is

$$U = \frac{1}{2} m \langle v^2 \rangle N \quad (4.11)$$

so its density is

$$u = \frac{1}{2} m \langle v^2 \rangle n \quad (4.12)$$

therefore we can insert this into the expression for pressure we have and find

$$P = \frac{2}{3} u \quad (4.13)$$

This is correct for mono-atomic gas such as hydrogen. In the more complicated case we can define the adiabatic index

$$\gamma_{ad} = \frac{C_p}{C_v} = 1 + \frac{2}{f} \quad (4.14)$$

where f is the number of degrees of freedom, and then we can find that

$$P = (\gamma_{ad} - 1)u \quad (4.15)$$

2.5 Generalizing To Radiation Pressure

The distribution of the velocities of the massive particles is the Boltzmann distribution. On the other hand, the distribution of the momenta of the photons is the Planck distribution:

$$\frac{dn}{d\nu} = n_\nu = \frac{8\pi\nu^2}{c^3} \frac{1}{e^{\frac{h\nu}{k_B T}} - 1} \quad (4.16)$$

and

$$u_\nu = n_\nu h\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{\frac{h\nu}{k_B T}} - 1} \quad (4.17)$$

To get the pressure, rather than doing an entirely new development, we can say our previous development of the pressure using a wall and particles was good it just needs some adjustments. We had:

$$P = \frac{1}{3}mn\langle v^2 \rangle = \langle \frac{1}{3}nv \cdot mv \rangle = \langle \frac{1}{3}nv \cdot p \rangle \quad (4.18)$$

inserting the momentum and velocity of photons we find

$$P = \langle \frac{1}{3}nc \frac{h\nu}{c} \rangle \quad (4.19)$$

$$= \langle \frac{1}{3}nh\nu \rangle \quad (4.20)$$

$$= \frac{1}{3}\langle u \rangle \quad (4.21)$$

$$= \frac{1}{3}u \quad (4.22)$$

And if we equate this to the more general pressure equation using the adiabatic constant we find

$$\gamma_{\text{photons}} = \frac{4}{3} \quad (4.23)$$

to rewrite this in terms of temperature we remember that

$$u = \int_0^\infty u_\nu d\nu = \int \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{\frac{h\nu}{k_B T}} - 1} \quad (4.24)$$

$$= [x = \frac{h\nu}{k_B T}] = 8\pi \frac{(kT)^4}{(hc)^3} \int_0^\infty x^3 \frac{dx}{e^x - 1} \quad (4.25)$$

$$= 8\pi \frac{(kT)^4}{(hc)^3} \cdot \frac{\pi^4}{15} \quad (4.26)$$

we can define a constant

$$a = \frac{8\pi^5 k_B^4}{15 (hc)^3} \quad (4.27)$$

and have

$$u = aT^4 \quad (4.28)$$

and find that the total pressure is

$$P_{tot} = nk_B T + \frac{1}{3}aT^4 \quad (4.29)$$

2.6 Nuclear Fusion

quick look at the periodic table. quick look at the standard model. the dominant nuclear fusion process is the pp-chain. The efficiency of the process is

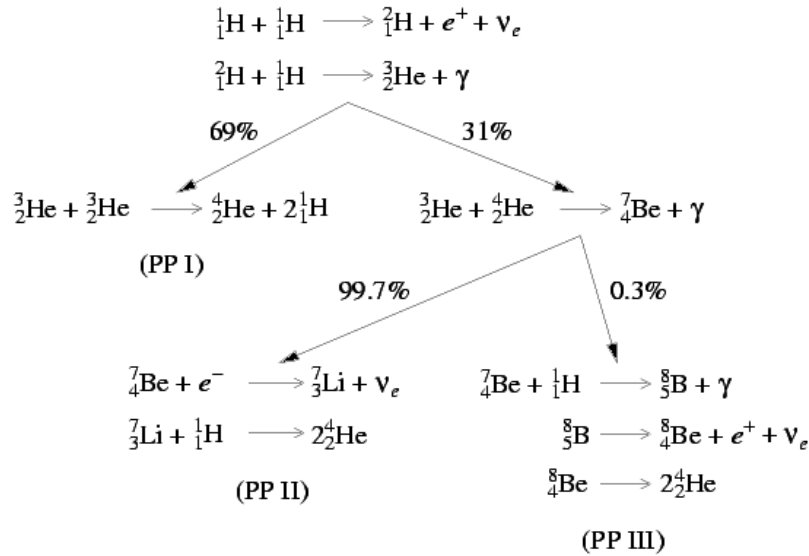


Figure 4.1: <https://burro.cwru.edu/academics/Astr221/StarPhys/ppchain.html>

$$\eta = \frac{E}{Mc^2} = \frac{26.7 \text{ MeV}}{4m_p c^2} = 0.7\% \quad (4.30)$$

for context of how much energy that is, we know that the sun's luminosity is $l_{\odot} = 4 \times 10^{33} \text{ erg/sec}$. This means that per second it produces $4 \times 10^{26} \text{ J}$, which is 10^6 times the yearly consumption of all humanity.

Lecture 5.

Thu, November 13th, 2025

To find the luminosity we need to also figure out the pace of the processes, which we'll call r_{xi} whose units are $\text{s}^{-1}\text{cm}^{-3}$, x indicating the target particle being hit by the particle i . First though let's check if nuclear fusion can happen in the sun at all. The electrostatic potential (in cgs) is given by

$$U = \frac{e^2}{r} = \frac{1.44 \text{ MeV}}{(r/\text{fm})} \quad (5.1)$$

where

$$e = 4.8 \times 10^{-10} \text{ esu} \quad \text{fm} = 10^{-13} \text{ m} \quad (5.2)$$

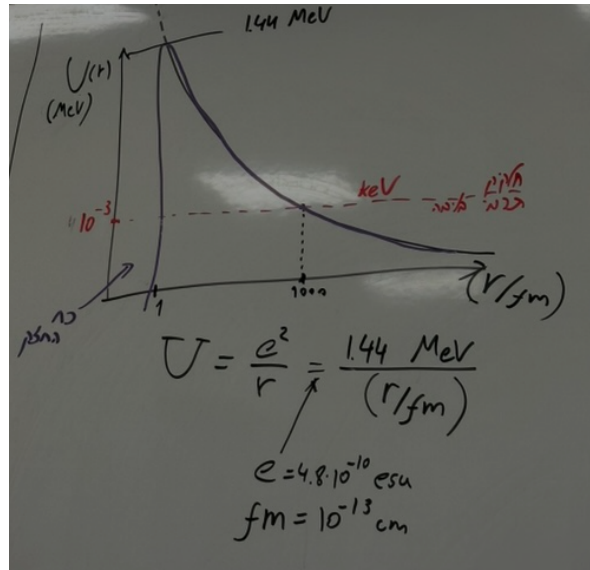
and the temperature at the core of the sun is $T_c \simeq 1 \times 10^7 \text{ K}$ and using the expression for the energy at the core

$$\frac{3}{2}k_B T = E \implies E_c \sim \text{keV} \quad (5.3)$$

This means that the thermal particles have only 0.1% of the energy required to jump the energy barrier imposed by the coulomb force's repulsion to reach the area the strong force attracts them. Seemingly if the system were purely classical, fusion could not happen. Let's consider an edge case where maybe fusion happens because a random small number of particles just happen to have a high enough energy to pass the barrier. To do that we can calculate how many particles on average there are that have an energy with approximately a mega-electronvolt of energy. For a rough estimate we can use statistical mechanics

$$e^{-\frac{E}{k_B T}} \sim e^{-1000} = 10^{-435} \quad (5.4)$$

where we inserted the temperature at the core of the sun and the energy we want to have to breach the barrier. The number is so small that there would on average be less than one particle in the entire universe that would have that energy. For that reason we didn't calculate it accurately, since it wouldn't have mattered. We'll need to take into consideration the effects of quantum tunneling, but before we do that, we need to discuss the action cross-section. We'll imagine a ball x with effective surface area σ being bombarded with particles i .



The number of particles hitting the area is given by

$$dN = \sigma v \delta t \cdot n_i \quad (5.5)$$

so the rate at which they hit is

$$\frac{dN}{dt} = \sigma v n_i \quad (5.6)$$

If we have an environment with many x particles then

$$r_{xi} = \sigma v n_i \cdot n_x \quad (5.7)$$

where n_x is the density of the particles x . In the more general case if we have a distribution of velocities then we have

$$v \Rightarrow v(E) \quad n_i \Rightarrow \frac{dn_i}{dE} dE \quad (5.8)$$

and then

$$r_{xi} = \int \sigma(E) v(E) n_x \frac{dn_i}{dE} dE \quad (5.9)$$

with the Stefan-Boltzmann law being

$$\frac{dn_i}{dE} = \frac{2n_i}{\sqrt{\pi}} \frac{\sqrt{E}}{(k_B T)^{3/2}} e^{-\frac{E}{k_B T}} \quad (5.10)$$

our goal now is to find $\sigma(E)$. For quantum tunneling we'll write

$$\sigma = \sigma_0 P \quad (5.11)$$

where P is the chance to tunnel and σ_0 is the effective area. As for σ_0 :

$$\sigma_0 \approx \pi R^2 \quad (5.12)$$

$$= \pi \lambda_{DB}^2 \quad (5.13)$$

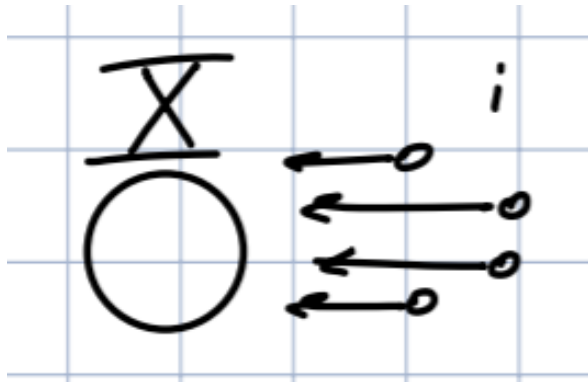
$$= \pi \left(\frac{h}{p} \right)^2 \quad (5.14)$$

$$= \pi \frac{h^2}{2\mu E} \quad (5.15)$$

where μ is the reduced mass, since we're working with a collision of two particles. or, in a general more accurate form

$$\sigma_0 = \frac{S}{E} \quad (5.16)$$

To find the chance to tunnel we'll refer to quantum mechanics but this isn't a course on quantum mechanics so it's just a quick overview.



The Schrödinger equation is

$$\frac{\hbar^2}{2\mu} \nabla^2 \Psi = (V(r) - E) \Psi \quad (5.17)$$

where our energy is

$$E = \frac{3}{2} k_B T = \frac{z_1 z_2 e^2}{r_1} \quad (5.18)$$

and our potential is

$$V(r) = \begin{cases} \frac{Z_A Z_B e^2}{r} & r > r_0 \\ 0 & r < r_0 \end{cases} \quad (5.19)$$

Solving for this potential is complicated. We'll approximate it using a step function. Its height should be the average of the potential

$$\langle V(r) \rangle = \frac{\int_{r_0}^{r_1} 4\pi r^2 V(r) dr}{\int_{r_0}^{r_1} 4\pi r^2 dr} = \frac{3}{2} E \quad (5.20)$$

and a step function is a problem we know how to solve. Inside the step we have

$$\frac{\hbar^2}{2\mu} \nabla^2 \Psi = \left(\frac{3}{2} E - E \right) \Psi \quad (5.21)$$

and since we're working in spherical coordinates

$$\frac{\hbar}{2\mu r} \frac{\partial^2}{\partial r^2} (r\Psi) = \frac{1}{2} E \Psi \quad (5.22)$$

whose solution is

$$\Psi = A \frac{e^{\beta r}}{r} \quad \beta = \frac{\sqrt{\mu E}}{\hbar} \quad (5.23)$$

and therefore the chance to find a particle in an area δr is proportional to $|\Psi| \delta r$ and the probability of tunneling is

$$P_{\text{tunnel}} = \frac{4\pi r_0^2 \delta r |\Psi(r_0)|^2}{4\pi r_1^2 \delta r |\Psi(r_1)|^2} = \frac{e^{2\beta r_0}}{e^{2\beta r_1}} = e^{2\beta(r_0 - r_1)} \approx e^{-2\beta r_1} \quad (5.24)$$

inserting the parameters we found earlier we find

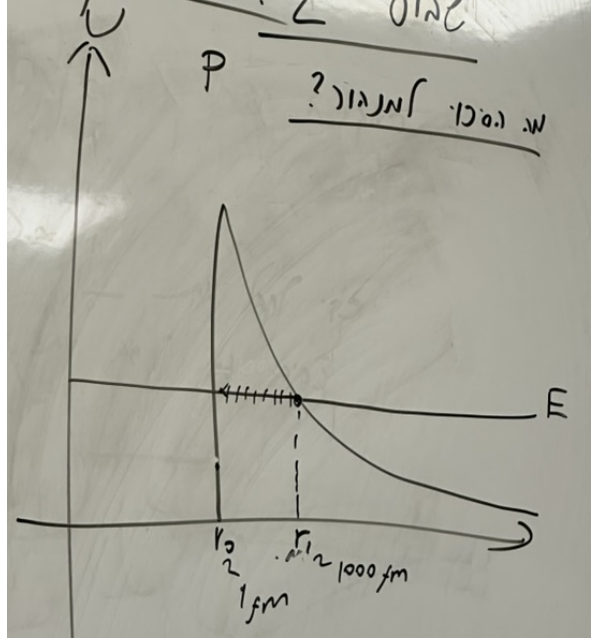
$$P \approx e^{-\frac{2\sqrt{\mu E}}{\hbar} \frac{Z_A Z_B e^2}{E}} \quad (5.25)$$

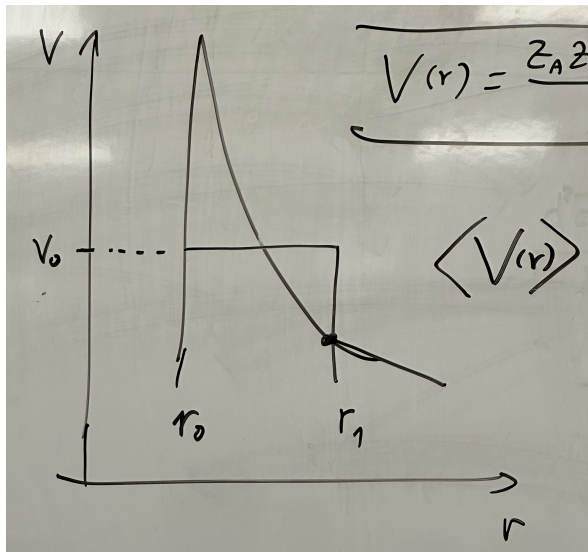
while the exact solution is

$$P = e^{-\sqrt{E_G/E}} \quad (5.26)$$

where

$$E_G = (\pi \alpha Z_A Z_B)^2 2\mu c^2 \quad \alpha = \frac{e^2}{\hbar c} \approx \frac{1}{137} \quad (5.27)$$





Lecture 6.

Mon, November 17th, 2025

It's worth noting that in the discussion on the probability of tunneling it is more likely the higher the energy, but the cross section for fusion becomes smaller, so it's a tradeoff. Specifically for the pp-chain the numbers are

$$Z_A = Z_B = 1 \qquad \mu = \frac{1}{2}m_p \qquad \mu c^2 = 0.94 \text{ GeV} \qquad (6.1)$$

giving us

$$E_G \approx 500 \text{ keV} \qquad (6.2)$$

therefore the probability of tunneling is

$$P = e^{-\sqrt{500}} = 2 \times 10^{-10} \qquad (6.3)$$

Let's check what this means for the sun. The number of particles in the sun's core is

$$N = 0.1 \frac{M_{\odot}}{m_p} \sim 10^{48} \qquad (6.4)$$

So even though the probability of tunneling is low, it definitely happens. To find the rate of the interactions we insert this into 5.9:

$$r_{xi} = \int \sigma n \frac{dn}{dv} dv \qquad (6.5)$$

$$= \int \sigma \sqrt{\frac{2E}{\mu}} \frac{dn}{dE} dE \qquad (6.6)$$

$$(6.7)$$

where

$$\frac{dn}{dE} = n_i \frac{2}{\sqrt{\pi}} \frac{\sqrt{E}}{(k_B T)^{3/2}} e^{-\frac{E}{k_B T}} \qquad (6.8)$$

$$\sigma = \frac{S}{E} e^{-\sqrt{E_G/E}} \qquad (6.9)$$

$$S = \frac{\pi \hbar^2}{2\mu} \qquad (6.10)$$

giving us

$$r_{xi} = n_i n_x \left(\frac{1}{k_B T} \right)^{3/2} \sqrt{\frac{8}{\pi \mu}} \int e^{-\left(E/K_B T + \sqrt{E_G/E} \right)} dE \qquad (6.11)$$

to figure out which particles are most responsible for the fusion process, the high energy ones or the low energy ones, if we designate

$$f(x) = e^{-\left(E/K_B T + \sqrt{E_G/E} \right)} \qquad (6.12)$$

then it looks like and the peak is found at

$$E_0 = \left(\frac{k_B T}{2} \right)^{2/3} E_G^{1/3} \qquad (6.13)$$

and specifically for the Sun

$$E_0 = 5 \text{ keV} \qquad (6.14)$$

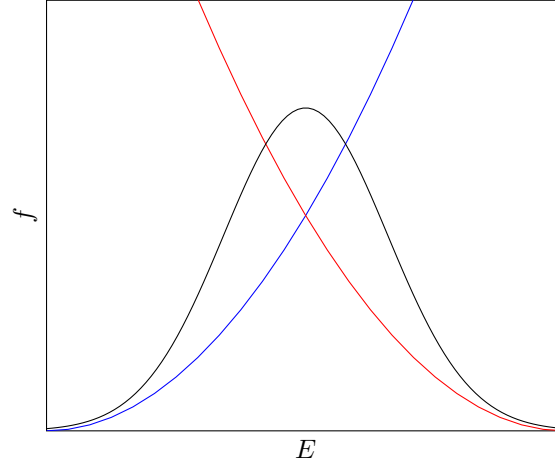


Figure 6.1: The portion of mass of the universe by element in the universe's early times.

Next we want to solve 6.11. It's not analytically solvable so we approximate it. What's often done in astrophysics is that we approximate things to a power law. Our function looks like this:

$$r_{xi} = An^2 f(T) \quad (6.15)$$

a power law approximation for this would look like

$$r_{xi} \approx An^2 T^\beta \quad (6.16)$$

Why do we do this? Often in astrophysics we look at relationships between values of parameters which span very large ranges, often entire orders of magnitude. for this reason instead of plotting regular linear graphs, we use log-log graphs. If we take a taylor series at some point and look at the first order it's a linear function. A linear line on a log-log graph fits a function of the form T^β . Back to the case of our Sun

$$r_{xi} = An_i n_x f(T) = A \rho_i \rho_x T^\beta \quad (6.17)$$

and if we define

$$x_i = \frac{\rho_i}{\rho} \quad x_x = \frac{\rho_x}{\rho} \quad (6.18)$$

then

$$r_{xi} = r_0 x_i x_x \left(\frac{\rho}{\rho_0} \right)^2 \left(\frac{T}{T_0} \right)^\beta \quad (6.19)$$

where r_0, ρ_0, T_0 are parameters we find from the fit. The amount of energy produced from all the fusion is

$$dE = E_0 r_{xi} dt dV \quad (6.20)$$

and inserting what we found so far

$$dE = E_0 r_0 x_i x_x \left(\frac{\rho}{\rho_0} \right)^2 \left(\frac{T}{T_0} \right)^\beta dV dt \quad (6.21)$$

which we are able to rewrite to an integral over the mass by using $dm = \rho dV$:

$$dE = \left(\frac{E_0 r_0}{\rho_0} \right) x_i x_x \left(\frac{\rho}{\rho_0} \right) \left(\frac{T}{T_0} \right)^\beta dm dt \quad (6.22)$$

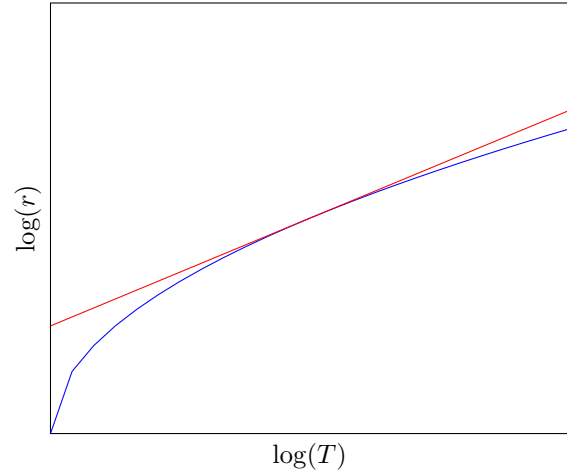


Figure 6.2: The portion of mass of the universe by element in the universe's early times.

where we can define

$$\epsilon_0 = \frac{E_0 r_0}{\rho_0} \quad (6.23)$$

which has units of energy produced per gram material per second, giving us

$$\frac{d^2 E}{dm^2} t = \epsilon_0 x_i x_x \frac{\rho}{\rho_0} \left(\frac{T}{T_0} \right)^\beta \quad (6.24)$$

and if we insert $\beta = 4$ which fits our Sun and

$$\epsilon_0 = 0.1 \text{ erg g}^{-1} \text{ s}^{-1} \quad \rho_0 = \text{g/cm}^3 \quad T_0 = 10^7 \text{ K} \quad x_i = x_x = 0.7 \quad (6.25)$$

then we find

$$\epsilon_{\text{O}} = 7.4 \text{ erg g}^{-1} \text{ s}^{-1} \quad (6.26)$$

then the luminosity produced is

$$dL = \epsilon dm = \epsilon \cdot 4\pi r^2 dr \rho \quad (6.27)$$

which gives us

$$\frac{dL}{dr} = \epsilon \cdot 4\pi r^2 \rho \quad (6.28)$$

2.7 Diffusion of Photons

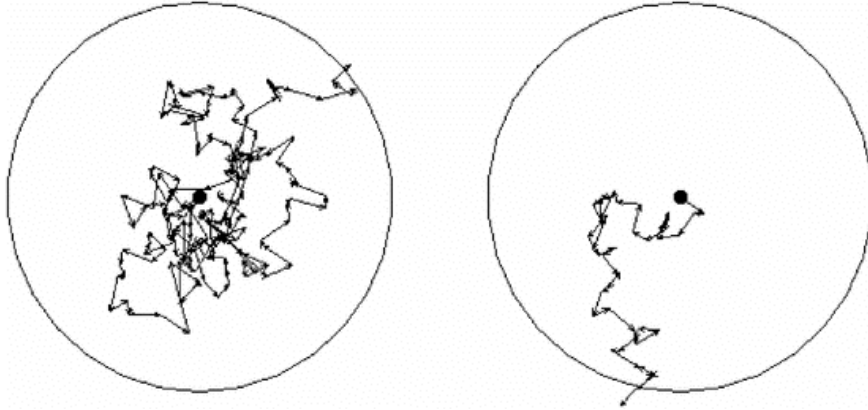


Figure 6.3: Example of Compton scattering Brownian motion (Random walk) of a photon.

Suppose we have a star and inside a photon is created in the middle and moves around inside until it reaches the outside. This process is called a random walk. Each step is of length λ and the star's radius is R .

$$R^2 = |\mathbf{R}|^2 = \left| \sum \mathbf{r}_i \right|^2 \quad (6.29)$$

$$= \sum_i \sum_j \mathbf{r}_i \cdot \mathbf{r}_j \quad (6.30)$$

$$= \sum_{i=1}^N |\mathbf{r}_i|^2 + \sum_{i \neq j} \mathbf{r}_i \cdot \mathbf{r}_j \quad (6.31)$$

$$= N\lambda^2 \quad (6.32)$$

therefore the number of steps until the photon escapes is

$$N = \left(\frac{R}{\lambda} \right)^2 \quad (6.33)$$

some typical numbers are

$$R = R_{\odot} \qquad \lambda \approx 0.1 \text{ cm} \qquad (6.34)$$

which gives us

$$N \approx 5 \times 10^{23} \qquad (6.35)$$

if we want the distance the photon will travel in total then

$$d = \lambda \cdot N \approx 4 \times 10^{22} \text{ cm} = 16 \text{ kpc} \qquad (6.36)$$

and the time it takes the photon to travel this distance is

$$t = d/c \approx 50\,000 \text{ yr} \qquad (6.37)$$

If want a more accurate description of the process, we can use the diffusion equation. We won't develop it from scratch but begin from Fick's Diffusion Law (A development from scratch can be found in Feynman's lectures volume 1 chapter 42):

$$\mathbf{J} = -D \nabla n \qquad (6.38)$$

where J is the flux of particles, $D = \frac{\lambda v}{3}$ is the diffusion parameter. We'll use this formula for photons. The free traversal is given by $\lambda = \frac{1}{\sigma n} = \frac{1}{\kappa \rho}$ where σ is the action cross section and $\kappa = \frac{\sigma}{m}$ is the opacity. So for photons we'll perform the substitutions

$$v \rightarrow c \qquad D \rightarrow \frac{\lambda c}{3} = \frac{c}{3\kappa\rho} \qquad \rho \rightarrow u_{\text{rad}} = aT^4 \qquad J \rightarrow f_{\text{rad}} = \frac{L_r}{4\pi r^2} \qquad (6.39)$$

which gives us

$$\frac{L_r}{4\pi r^2} = f_{\text{rad}} = -\frac{1}{3} \frac{c}{\kappa\rho} \frac{d}{dr} aT^4 \qquad (6.40)$$

rearranging differentiating we find

$$\frac{dT}{dr} = -\frac{3}{4ac} \frac{\kappa\rho}{T^3} \frac{L_r}{4\pi r^2} \qquad (6.41)$$

In conclusion we have four differential equations:

$$\frac{dP}{dr} = -G \frac{M_r \rho}{r^2} \qquad \frac{dM_r}{dr} = 4\pi r^2 \rho \qquad (6.42)$$

$$\frac{dL_r}{dr} = 4\pi r^2 \rho \epsilon \qquad \frac{dT}{dr} = -\frac{3}{4ac} \frac{\kappa\rho}{T^3} \frac{L_r}{4\pi r^2} \qquad (6.43)$$

together with the state equation P , the energy production equation ϵ , and the opacity κ we have all the tools we need to describe the structure of stars.

Lecture 7.

Thu, November 20th, 2025

For an approximate set of solutions we can assume that

$$T, P, \rho|_{r=R_*} = 0 \quad M_r = L_r|_{r=0} = 0 \quad \begin{cases} M_r|_{R_*} = M \\ L_r|_{R_*} = L \end{cases} \quad (7.1)$$

In the process of diffusion, photons can undergo several kinds of interactions

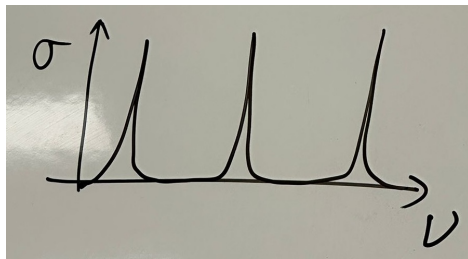
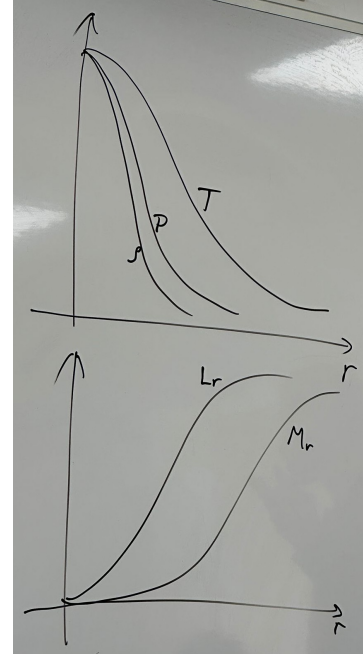
Bound-bound (Excitation) $X + \nu \rightarrow X^*$.

Bound-free (Ionization) $X^n + \nu \rightarrow X^{n+1} + e$.

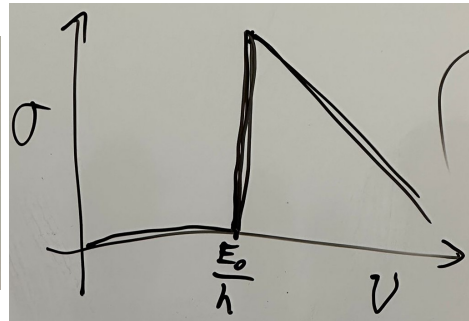
With $\sigma = \sigma_0 \left(\frac{h\nu}{E_0} \right)^{-3}$ for hydrogen and atoms with one electrons.

Free-free absorption (Reverse bremsstrahlung)

Free-free e-scattering (Thomson) which has a well known action cross section $\sigma_T = \frac{8}{3}\pi r_e^2$ where $r_e = \frac{e^2}{m_e c^2}$. Additionally, if there are ions in the area then if $\lambda \ll r_p$ there can happen Compton scattering and if $\lambda \gg r_p$ there can happen Rayleigh scattering.



(a) Bound-bound (excitation)



(b) Bound-free (Ionization)

Mon, November 24th, 2025

3 L-M ties

Looking at the HR diagram we see that there appears to be a connection between mass and luminosity. We'll dwell on this for a while. This also connects to its life span and evolution. Observationally, if we plot the luminosity as a function of the mass, we find something that looks like this

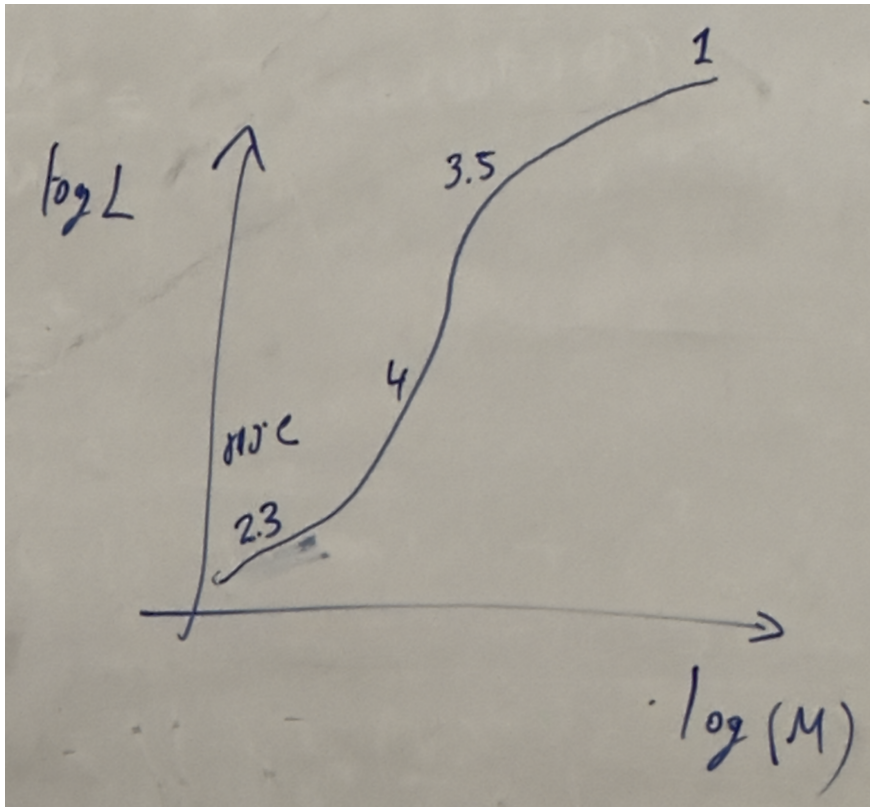


Figure 8.1: A diagram of the L-M relation

and using the relation

$$\frac{L}{L_{\odot}} = A \left(\frac{M}{M_{\odot}} \right)^{\alpha} \quad (8.1)$$

we can write this as a table To get a feeling for why this behaves like this, we'll do the following.

$M/M_{\odot}$	A	α
< 0.43	0.23	2.3
0.43–2	1	4
2–55	1.4	3.5
> 55	3.2×10^4	1

We'll assume

$$\kappa = \kappa_{\text{Thomson}} = \text{const.} \quad (8.2)$$

which is a good approximation for massive stars (2 to 55 solar masses). We'll also assume a power law of the form

$$M_r \propto r^\alpha \quad P \propto r^\beta \quad T \propto T^r \quad (8.3)$$

therefore the stellar structure equations become

$$\beta \frac{P}{r} \propto \frac{M\rho}{r^2} \quad \alpha \frac{M}{r} \propto r^2 \rho \quad \gamma \frac{T}{r} \propto \frac{L\kappa\rho}{r^2 T^3} \quad (8.4)$$

rearranging we find

$$P \propto \frac{M\rho}{r} \quad M \propto r^3 \rho \quad L \propto \frac{T^4 r}{\rho} \quad P \propto \rho T \quad (8.5)$$

Inserting D into A we find

$$T \propto \frac{M}{r} \quad (8.6)$$

and inserting that into C

$$L \propto \frac{(M/r)^4 r}{\rho} = \frac{M^4/r^3}{\rho} \propto M^3 \quad (8.7)$$

For extremely massive stars ($> 55M_\odot$), the radiation pressure is dominant, at which point D no longer holds and instead

$$P \propto T^4 \quad (8.8)$$

which inserted into A gives

$$T^4 \propto \frac{M\rho}{r} \quad (8.9)$$

and inserting that into C

$$L \propto \frac{(M\rho/r)r}{\rho} = M \quad (8.10)$$

3.1 The Most Massive Stars

Let's look at the outer layer of a very massive star. Using the radiation propagation equation

$$\frac{dT}{dr} = -\frac{3}{4ac} \frac{\kappa\rho}{T^3} \frac{L_r}{4\pi r^2 c} \quad (8.11)$$

we can rearrange it to

$$\frac{1}{3} \frac{d}{dr} (aT^4) = -\kappa\rho \frac{L_r}{4\pi r^2 c} \quad (8.12)$$

and remembering that $\frac{1}{3}aT^4$ is the radiation pressure

$$\frac{dP_{\text{rad}}}{dr} = -\kappa\rho \frac{L_r}{4\pi r^2 c} \quad (8.13)$$

on the other hand, from the hydrostatic equations we have

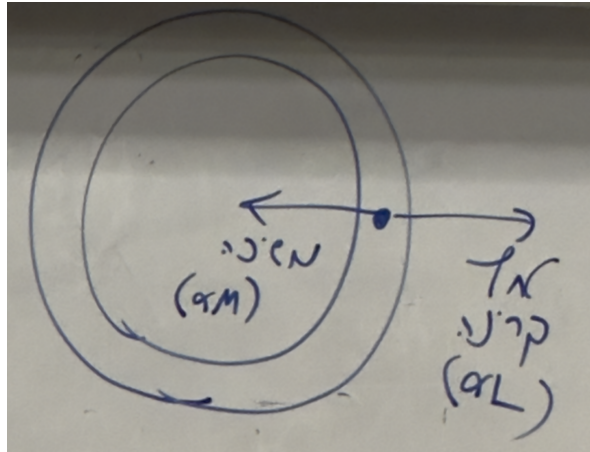
$$\frac{dP}{dr} = -\frac{GM_r\rho}{r^2 c} \quad (8.14)$$

setting our two expression to equal we find

$$\frac{GM_r\rho}{r^2} = \kappa\rho \frac{L_r}{4\pi r^2 c} \quad (8.15)$$

isolating L_r and inserting $r = R$

$$L_r = 4\pi c \frac{GM}{\kappa} \quad (8.16)$$



this expression for luminosity is often called Eddington Luminosity. This expression gives us the maximum luminosity a star can have before it becomes unstable. If for a given star's parameters the observed luminosity is higher than the Eddington luminosity, then the radiation pressure will begin to overpower its hydrostatic pressure which will destabilize the star. Let's plug some numbers in to get a feel. Inserting the constants we have

$$L = 3.8 \times 10^4 \frac{M}{M_{\odot}} L_{\odot} \quad (8.17)$$

and for a mass of $M = 60M_{\odot}$

$$L = 2.3 \times 10^6 L_{\odot} \quad (8.18)$$

3.2 The Lifespan Of Stars

For the Sun we can write

$$t_{\odot} \simeq \frac{\eta M_{\text{core}} c^2}{L_{\odot}} \sim 10 \text{ Gyr} \quad (8.19)$$

If we look at stars with different masses then

$$t_* \propto \frac{M}{L} \propto M^{-3} \implies t_* = t_{\odot} \left(\frac{M}{M_{\odot}} \right)^{-3} \quad (8.20)$$

3.3 Convection

Convection becomes important if

$$t_{\text{rad}} \gg t_{\text{dyn}} = \frac{l}{c_s} \quad (8.21)$$

A more concrete criteria for convection

Let's look at a element of gas with density ρ and temperature T at radius r in a star, which moves to $r + dr$ and its parameters change to $\rho + \delta\rho$ and $T + \delta T$. The criterion for convection is

$$\rho_{\text{element}} + \delta\rho_{\text{element}} < \rho_{\text{env}} + \delta\rho_{\text{env}} \quad (8.22)$$

or

$$\delta\rho < d\rho \quad (8.23)$$

then the element will float upwards. For a more accurate derivation we'll assume this is an adiabatic process, which lets us say

$$P \propto \rho^{\gamma} \quad \frac{\delta\rho}{\rho} = \frac{1}{\gamma} \frac{\delta P}{P} = \frac{1}{\gamma} \frac{dP}{P} \quad (8.24)$$

we'll also assume there is a pressure equilibrium with the star. Therefore

$$P = \frac{\rho}{m} \kappa T \quad (8.25)$$

which gives us

$$\frac{d\rho}{\rho} = \frac{dP}{P} - \frac{dT}{T} \quad (8.26)$$

We can now insert both 8.24 and 8.26 into 8.23 we find

$$\frac{1}{\gamma} \frac{1}{P} \frac{dP}{dr} < \frac{1}{P} \frac{dP}{dr} - \frac{1}{T} \frac{dT}{dr} \quad (8.27)$$

rearranging

$$\frac{dT}{dr} < \frac{\gamma - 1}{\gamma} \frac{T}{P} \frac{dP}{dr} \quad (8.28)$$

both the differentials are smaller than 0, so we'll multiply both sides by -1 and find

$$\left| \frac{dT}{dr} \right| > \frac{\gamma - 1}{\gamma} \frac{T}{P} \left| \frac{dP}{dr} \right| \quad (8.29)$$

The effect this has is that it improves the transfer of heat which in turn balances it out recursively, and after a while the greater symbol will become equal

$$\left| \frac{dT}{dr} \right| = \frac{\gamma - 1}{\gamma} \frac{T}{P} \left| \frac{dP}{dr} \right| \quad (8.30)$$

which is a stable state. If 8.29 is satisfied, we replace the radiation diffusion equation for the structure of stars with the convection equation 8.30.

Lecture 9.

Thu, November 27th, 2025

I missed this lecture. The presentation “Supernova, Cosmic-rays and Turbulence ” was covered in this lecture.

Lecture 10.

Mon, December 1st, 2025

I missed this lecture.

Thu, December 4th, 2025

I missed this lecture. This lecture and the previous one covered pages 61 to 80 in Shmuel's handwritten notes. The notes I'm including here for this lecture are copied from those notes as closely to word for word as I could, and are not based on the lecture itself.

4 Remnants of Stars - Exotic Objects

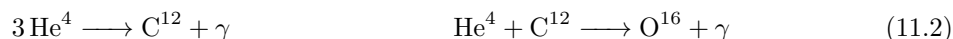
Stars on the main sequence towards the end of their lives will become either Red Giants (If their mass is below 8 solar masses) or Supergiants (If their mass is above 8 solar masses). Red Giants then become Planetary Nebulas and eventually White Dwarfs. Supergiants collapse and explode in a Supernova and then end up as either a Neutron Star or a Black Hole.

4.1 White Dwarfs

In the 19-20 centuries it was determined that the nearby star Sirius (the brightest star in the night sky) is actually a pair of two stars, Sirius A and B, which is a white dwarf. From Kepler's law it was determined that Sirius B's mass is approximately 1 solar mass, but because it had low luminosity and high temperature, it was concluded that its radius is approximately 6000 km. This means its average density is

$$\bar{\rho} = \frac{M}{\frac{4\pi}{3}R^3} = 2 \times 10^6 \text{ g/cm}^3 = 2 \text{ ton/cm}^3 \quad (11.1)$$

in other words, a teaspoon of a white dwarf is as heavy as an elephant. At the end of the star's life the core shrinks (because the fusion of hydrogen stops). Therefore both the density and temperature of the core grow, and as a result Helium and Carbon begin to fuse in the processes



the first process being called "triple alpha" which releases 7.275 MeV. Eventually the Helium is used up which further compresses and heats the core, but eventually something stops the shrinking. Pauli's exclusion principle forbids electrons from coming too close together.

Which density does this start at?

When the distance between electrons become of the order of magnitude of $\frac{1}{2}\lambda_{DB}$ the wavefunctions start to 'feel' each other. In terms of density,

$$\rho_q \simeq \frac{m_p}{(\frac{1}{2}\lambda_{DB})^3} = \frac{8m_p(3m_e k_B T)^{3/2}}{h^3} = 10^4 T_8^{3/2} \text{ g/cm}^3 \quad (11.3)$$

where $T_8 \equiv \frac{T}{10^8 \text{ K}}$. Once $\rho \gtrsim \rho_q$ the 'magic' happens. For the core of the sun, $T_8 = 0.015$ which gives $\rho_q = 640 \text{ g/cm}^3$ but since the sun's core's density is 150 g/cm^3 it is very far from the quantum limit. But when a star's core shrinks it is possible to reach it.

At The Quantum Limit

Particles' energies are distributed either by the Bose-Einstein distribution (for photons, gluons, etc) or the Fermi-Dirac distribution (for electrons, protons, neutrons, etc). The FD distribution at temperatures $k_B T \ll \epsilon_f$ it is approximately a step function where all energies $\epsilon \leq \epsilon_f$ are populated and the rest are not. The fermi energy is

$$\epsilon_f = \frac{\hbar^2}{2m_e} (3\pi^2 n_e)^{2/3} \quad (11.4)$$

The lecture notes here develop the fermi energy, I skip it as it's included in the prerequisite course *Statistical Mechanics*. In conclusion, we'll find particles with high energy and momentum regardless of the temperature. They're being forced to be such by Pauli's exclusion principle.

The Pressure

Earlier in the course we developed the equation

$$P = \frac{1}{3} \int_0^{\infty} dp p v \left(\frac{dn}{dp} \right) \quad (11.5)$$

if we assume the particles are non-relativistic we can substitute the velocity

$$P_e = \frac{1}{3m_e} \int_0^{\infty} dp p^2 \left(\frac{dn_e}{dp} \right) \quad (11.6)$$

the number of electrons as a function of their momentum is

$$n_e = \frac{8\pi}{3} \left(\frac{p}{h} \right)^3 \quad (11.7)$$

so the density is

$$\frac{dn_e}{dp} = \begin{cases} 8\pi \frac{p^2}{h^3} & p \leq p_f \\ 0 & p > p_f \end{cases} \quad (11.8)$$

inserting this back into 11.6

$$P_e = \frac{1}{3m_e} \int_0^{p_f} dp p^2 \cdot 8\pi \frac{p^2}{h^3} \quad (11.9)$$

$$= \frac{8\pi}{3m_e h^3} \int_0^{p_f} dp p^4 = \frac{8\pi}{15m_e h^3} p_f^5 \quad (11.10)$$

inserting p_f

$$P_e = \left(\frac{3}{8\pi} \right)^{2/3} \frac{h^2}{5m_e} n_e^{5/3} \quad (11.11)$$

To find the expression in terms of density of mass we notice that

$$\rho = n_+ \cdot Am_p \quad (11.12)$$

and if we assume the material is fully ionized we can write $n_e = n_+ \cdot Z$ which gives us

$$\rho = n_+ Am_p = \frac{n_e}{Z} Am_p = m_p n_e \frac{A}{Z} \quad (11.13)$$

Therefore

$$n_e = \frac{\rho}{m_p} \frac{Z}{A} \quad (11.14)$$

inserting this into 11.9

$$P_e = \left(\frac{3}{\pi} \right)^{2/3} \frac{h^2}{20m_e m_p^{5/3}} \left(\frac{Z}{A} \right)^{5/3} \rho^{5/3} \quad (11.15)$$

For a typical White Dwarf, $P_e \approx 3 \times 10^{22}$ dyne/cm² which is 5 orders of magnitude more than the sun's core.

An Estimate For a Typical Radius

Using the hydrostatic equation

$$\frac{dP}{dr} = -\frac{GM_r}{r^2}\rho \quad (11.16)$$

and approximating

$$P \propto r^\beta \implies \frac{dP}{dr} \sim \frac{P}{r} \implies P \approx \frac{GM_\rho}{r} \quad (11.17)$$

$$\rho \approx \frac{M}{r^3} \implies \approx \frac{GM^2}{r^2} \quad (11.18)$$

Some development leads to

$$r_{WD} = \frac{h^2}{20m_em_p^{5/3}G} \left(\frac{Z}{A}\right)^{5/3} M^{-1/3} \quad (11.19)$$

inserting constants

$$r_{WD} \approx 10^9 \text{ cm} \cdot \left(\frac{Z}{A}\right) \left(\frac{M}{M_\odot}\right)^{-1/3} \quad (11.20)$$

What is P_e at the relativistic limit?

Inserting $v = c$ into 11.5 we find

$$P_e = \left(\frac{3}{8\pi}\right)^{1/3} \frac{hc}{4m_p^{4/3}} \left(\frac{Z}{A}\right)^{4/3} \rho^{4/3} \quad (11.21)$$

to find the mass-radius relation in the relativistic case we'll equate the hydrostatic pressure to the degeneracy pressure and find

$$\frac{GM^2}{r^4} \approx d\rho^{4/3} \quad (11.22)$$

if we insert the full expression for the density we'll find that the radius cancels out! This means we can't have an expression to connect the radius and mass! We find only a condition on the mass:

$$M_{ch} = 0.21 \left(\frac{hc}{Gm_p}\right)^{3/2} \left(\frac{Z}{A}\right)^2 m_p \quad (11.23)$$

This is called the Chandrasekhar mass, and it is the maximum allowed mass for a white dwarf. If $M > M_{ch}$ the degeneracy pressure doesn't hold it together and it will collapse further. It's interesting to note that the Chandrasekhar mass has constants from different fields of physics.

Cooling of White Dwarfs

When White Dwarfs are created they are very hot ($T \sim 10^9 \text{ K}$), and they cool down through black body radiation. To find the time it takes to cool down we can equate the luminosity of the star to the time derivative of the ideal gas law

$$-\sigma T^4 \cdot 4\pi r^2 = \frac{M}{m_p} k_B \frac{dT}{dt} \quad (11.24)$$

Therefore

$$t_{cool} = \int_0^{t_{cool}} dt \quad (11.25)$$

$$= \frac{(M/m_p) k_B}{4\pi r^2 \sigma} \int_{T_0}^{T_f} \frac{dT}{T^4} \quad (11.26)$$

$$= -\frac{(M/m_p) k_B}{4\pi r^2 \sigma} \left(-\frac{1}{3}\right) \left(\frac{1}{T_f^3} - \frac{1}{T_0^3}\right) \quad (11.27)$$

setting the initial inverse temperature cubed to be negligible we find

$$t_{cool} = \frac{(M/m_p) k_B}{12\pi r^2 \sigma} \frac{1}{T_f^3} \quad (11.28)$$

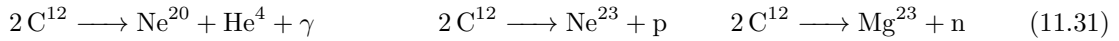
so the time it would take a white dwarf to cool to 1000 kelvin would be

$$t_{cool} = 15 \text{ Gyr} \left(\frac{M}{M_\odot}\right) \left(\frac{4}{4000 \text{ km}}\right)^{-2} \quad (11.29)$$

which means it'd take longer than the age of the universe. In practice the cooling time is even longer. For this reason white dwarfs appear white (or teal).

4.2 What Happens To More Massive Stars ($M > 8M_\odot$)?

After the fusion of hydrogen ends, the pressure goes down, their core shrinks, it gets hotter, and fusion of the next element begins. First the triple alpha reaction, until it runs out too and the process happens again. The next stage is fusion of Carbon into Oxygen and Oxygen into Neon



and so on. Each stage is shorter than the last this continues until it reaches Iron. From this point

H	5 Myr
He	0.5 Myr
C	500 yr
Ne	1 yr
Si	1 day

fusion no longer takes place, which destabilizes the core. There is no source of pressure which causes the core to collapse and heat up. This causes two processes to take place, one of which is called photodisintegration, which is the process



the first process consumes 124 MeV and the second 28 MeV. If we assume a white dwarf converted all its mass to iron the number of iron atoms is

$$N_{Fe} = \frac{M_{ch}}{m_{Fe}} \sim 3 \times 10^{55} \quad (11.33)$$

So the total energy to be lost is

$$E_{loss} \sim 2 \times 10^{52} \text{ erg} \quad (11.34)$$

which is a massive amount of energy, equivalent to the total energy radiated by the sun through its entire life.

Lecture 12.

Mon, December 8th, 2025

After both processes our star's material is rich in neutrons, which are fermions, and therefore Pauli's exclusion principle applies to them too. Everything is the same as for elections, the only difference being that they have a different λ_{DB} . A similar analysis to electrons can be done with two differences: using the neutron mass instead of the electron's, and setting $Z/A = 1$. This results in

$$R_{ns} = 2.3 \times 10^9 \frac{m_e}{m_n} \left(\frac{M}{M_\odot} \right)^{-1/3} = 11 \text{ km} \left(\frac{M}{M_\odot} \right)^{-1/3} \quad (12.1)$$

which is about 500 times smaller than a white dwarf. The density in comparison to a white dwarf will be

$$\frac{\rho_{ns}}{\rho_{wd}} \sim 500^3 \sim 10^8 \implies \rho_{ns} \sim 10^{14} \text{ g/cm}^3 \quad (12.2)$$

If we look at the density of material in the nucleon of an atom it is

$$\rho_{nuc} = \frac{m_p}{\frac{4\pi}{3} (1 \text{ fm})^3} \sim 4 \times 10^{14} \text{ g/cm}^3 \quad (12.3)$$

So the density of a neutron star is in comparable to that of the nucleus of an atom. You could think of the entire thing as a single atom nucleus with atomic number 10^{57} . Recalling the expression for M_{ch} :

$$M_{ch} = 0.21 \left(\frac{Z}{A} \right)^2 \alpha_G^{-3/2} m_p \quad (12.4)$$

setting the relation we had for neutron stars

$$M_{ns} = 0.21 \alpha_G^{-3/2} m_p = 5.6 M_\odot \quad (12.5)$$

which is the upper limit for the mass of a neutron star. A neutron star more massive than that won't hold itself together and collapse further. Observations indeed haven't found any neutron stars heavier than (or even close to) that mass. More accurate calculations give a result significantly lower. The reason for this is that general relativistic effects become important at this scale. Relativistic effects become relevant when

$$\frac{E_{Gr}}{mc^2} \sim \frac{\frac{GM_{ns}m}{R_{ns}}}{mc^2} = \frac{GM_{ns}}{R_{ns}c^2} \sim 0.2 \quad (12.6)$$

Another important effect we haven't mentioned yet is the strong nuclear force. Since the neutron star effectively behaves like a giant nucleus, the strong force will start to have an effect. We don't quite understand what its effect is though. If we want to look at how much energy is expected to be released in the collapse of a core into a neutron star

$$\Delta E_{gr} = \left| \frac{\frac{3}{5}GM^2}{R_0} - \frac{\frac{3}{5}GM^2}{R_{ns}} \right| \approx \frac{\frac{3}{5}GM^2}{R_{ns}} \quad (12.7)$$

and if we insert typical quantities for neutron stars

$$M = 1.3 M_\odot \quad R_{ns} = 11 \text{ km} \quad (12.8)$$

we find

$$\Delta E_{gr} \sim 3 \times 10^{53} \text{ erg} \quad (12.9)$$

This energy has to go somewhere and this causes a supernova. This is the leading process which causes supernovas. This kind of supernova is called a core-collapse supernova. The luminosity we see from this process is

and the total energy we observe is

$$E_\gamma = \int dt L = 3 \times 10^{49} \text{ erg} \quad (12.10)$$

which is only 0.01% of the available energy. Where does the rest of the energy go? Maybe it is ejected from the supernova. The ejected mass is about $(1 - 10) M_\odot$ and observations show that its velocity is $v \sim 10^4 \text{ km s}^{-1}$. This gives us an energy of

$$E_k \sim \frac{1}{2} M_{ej} v^2 = 3 \times 10^{51} \text{ erg} \quad (12.11)$$

which is 1% of the total energy, so there must be another explanation. Approximately 99% is ejected as neutrinos. The density in the center of the supernova is extremely high and the temperature is very high, which means there are very many photons whizzing around close to each other, and some of them collide with each other which results in

$$\gamma + \gamma \rightarrow e + e^+ \rightarrow \begin{cases} \nu_e + \bar{\nu}_e \\ \nu_\mu + \bar{\nu}_\mu \\ \nu_\tau + \bar{\nu}_\tau \end{cases} \quad (12.12)$$

The conditions are so extreme that sometimes even the neutrinos go through collision processes. The reason this process doesn't happen in the sun is that there just isn't enough energy there. The rest energy of the electron is $m_e c^2 = 0.5 \text{ MeV}$ which photon collisions can't reach in the sun. Other types of explosions: Type 1a Supernovas. There's a white dwarf just chilling on its own but a red giant approaches it and the white dwarf starts sucking material out of the red giant. It then eventually reaches M_{ch} and explodes. In regular stars there is a negative feedback loop where fusion causes temperature to increase, which increases the pressure, which increases the radius of the star, which lowers the temperature, which lower the rate of fusion. In contrast in a white dwarf fusion causes an increase in temperature which doesn't affect the pressure, so it increases the rate of fusion again, and so on. This is called a type 1a supernova. Another kind of explosion is a gamma ray burst. The energy released from GRBs is $E \sim 10^{51} \text{ erg}$ which is all released over about 0.1 seconds to a minute. GRBs are an active research area and it's theorized that there are several types.

Mon, December 15th, 2025

4.3 Pulsars

Pulsars are supernova remnants which appear to have pulsing fluxes of radiation. For examples, at the center of the Crab Nebula there is a pulsar which pulses with every 33 milliseconds. The current theory is that these pulses come from neutron stars with magnetic fields which rotate very rapidly. These magnetic fields create jets that go in opposite directions. We only observe these as pulsars if the jets point at us.

Why do they spin so fast?

If we look at a typical neutron star of radius 10 km, and if it spins with period of 1 ms, then the speed of its surface is as fast as $0.2c$. Where does that speed come from? Conservation of angular momentum. Angular momentum for a sphere is

$$J = I\omega = \frac{2}{5}MR^2\omega \quad (13.1)$$

to satisfy conservation of angular momentum we have

$$J_0 = J_1 \quad (13.2)$$

$$\eta MR_1^2\omega_1 = \eta MR_0^2\omega_0 \quad (13.3)$$

$$\implies \frac{\omega_1}{\omega_0} = \left(\frac{R_1}{R_0}\right)^{-2} \quad (13.4)$$

for example, plugging in the sun's radius for the initial and a neutron star radius for the final radius, then we have

$$\left(\frac{R_1}{R_0}\right)^{-2} = 5 \times 10^9 \quad (13.5)$$

and since the sun's rotation's period is $\tau_0 = 25 \text{ d} = 2 \times 10^6 \text{ s}$, we have

$$\tau_1 = \tau_0/5 \times 10^9 \sim \text{ms} \quad (13.6)$$

Where does the magnetic field come from?

The magnetic field of a pulsar is of approximately 10^{13} G . In comparison, the sun's magnetic field strength is 1 G , and the earth's is 10^{-5} G . We can think of a pulsar as a rotating magnetic dipole. The radiation from this dipole is given by

$$L = \frac{1}{6c^3} B^2 R^6 \omega^4 \sin^2 \theta \quad (13.7)$$

For the pulsar in the crab nebula we measure

$$L = 5 \times 10^{38} \text{ erg s}^{-1} \quad (13.8)$$

$$\omega = 190 \text{ s}^{-1} \quad (13.9)$$

and if we assume a typical 10 km radius, we find $B \approx 10^{13} \text{ G}$.

How does the magnetic field get so strong?

During the collapse process, another process called flux freezing takes place. The magnetic flux is defined by, for stars

$$\Phi_B \propto BA = B \cdot \pi R^2 \quad (13.10)$$

doing a similar conservation of flux calculation we find

$$B_0 R_0^2 = B_1 R_1^2 \quad (13.11)$$

$$B_1 = B_0 \left(\frac{R_1}{R_0} \right)^{-2} \quad (13.12)$$

which fits our comparison earlier to the sun, if we plug in the numbers.

Observation of the Crab Nebula

Observation of the crab nebulas show that the frequency of pulses from the pulsar in the crab nebula today is

$$\omega_0 = 190 \text{ s}^{-1} \quad (13.13)$$

and the rate of change in the frequency is

$$\dot{\omega} = 2.4 \times 10^{-9} \text{ s}^{-2} \quad (13.14)$$

so it's slowing down. To see why it's slowing down, we'll look at the energy it loses through radiation

$$\frac{dE}{dt} \propto \omega^4 \quad (13.15)$$

on the other hand

$$E = I\omega^2 \implies \frac{dE}{dt} = \omega \dot{\omega} \quad (13.16)$$

equating the two we find

$$\dot{\omega} = c\omega^3 \quad (13.17)$$

after solving we find

$$t = \frac{1}{2c} \left(\frac{1}{\omega_o^2} - \frac{1}{\omega_i^2} \right) \quad (13.18)$$

and with a little rearranging we have

$$t = \frac{\omega_0^3}{\dot{\omega}_0} \left(\frac{1}{\omega_0^2} - \frac{1}{\omega_i^2} \right) \quad (13.19)$$

if we assume significant deceleration $\omega_0 \ll \omega_i$

$$t = \frac{\omega_0}{\dot{\omega}_0} \quad (13.20)$$

inserting the numbers we observed we find

$$t = 1260 \text{ yr} \quad (13.21)$$

4.4 Black Holes

If the mass of the star is greater than the maximum mass allowed for neutron stars it becomes a black hole. This gives us a lot of mass concentrated in a very small radius.

What is the escape velocity?

The kinetic energy of an object is

$$E_k = \frac{1}{2}mv^2 \quad (13.22)$$

while the gravitational potential energy is

$$E_G = \frac{GMm}{R} \quad (13.23)$$

equating the two we find

$$v_{esc} = \sqrt{\frac{2GM}{R}} \quad (13.24)$$

if we equate this to the speed of light we find

$$R_s = \frac{2GM}{c^2} \quad (13.25)$$

which means that if R, R_s even light cannot escape the star's surface. This is called a black hole. Some examples for Schwartzchild radii are this calculation is technically incorrect because we ignored

R_s	M
0.9 cm	$M_{\oplus} = 6 \times 10^{27} \text{ g}$
3 km	$M_{\odot} = 2 \times 10^{33} \text{ g}$
0.08 au	$M = 4 \times 10^6 M_{\odot}$
8 au	$4 \times 10^8 M_{\odot}$

the effects of relativity. The results are correct mostly by accident.

5 General Relativity

The Einstein equation is

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (13.26)$$

where $T_{\mu\nu}$ is the momentum-energy tensor and it describes the 'sources' and $G_{\mu\nu}$ describes the curvature of spacetime. $G_{\mu\nu}$ is comprised of the metric $g_{\mu\nu}$ and its derivatives. The simplest metric is the Minkowski metric which describes flat spacetime. We work with mostly minus, therefore the metric looks like

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (13.27)$$

and the four-position is defined by

$$dx^\mu = \begin{cases} dx^0 = c dt \\ dx^1 = dx \\ dx^2 = dy \\ dx^3 = dz \end{cases} \quad (13.28)$$

As we know many quantities aren't invariant to lorentz boosts, e.g lengths and time. But there are some quantities which are invariant. An important lorentz invariant is the 'interval' ds :

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \quad (13.29)$$

this value can be either positive, negative, or zero. We can look at this through a figure. the volume inside the cone represents events which are causally linked. Massive objects could have moved between those two events, and for those events the interval is positive and a different frame of refence can be found in which both events happen at the same location. These events are called timelike. There are also events outside the cone. These events are not causally linked so no information can reach

them. For them the interval is negative. A different frame of reference can be found in which both events happen at the same time. These events are called spacelike. The final kind of events are those directly on the cone. These events are called lightlike and for these events they are causally linked only for massless particles and the interval is equal 0.

5.1 The Minkowski Metric in Spherical Coordinates

We know what the metric looks like in cartesian coordinates, and we know the transformation to spherical coordinates, therefore

$$ds^2 = (c dt)^2 - dr^2 - (r d\theta)^2 - (r \sin \theta d\varphi)^2 \quad (13.30)$$

therefore the metric is

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -r^2 & 0 \\ 0 & 0 & 0 & -r^2 \sin^2 \theta \end{pmatrix} \quad (13.31)$$

Lecture 14.

Mon, December 22nd, 2025

If our space does have a mass M at the position $r = 0$ then the metric for the space is the Schwartzchild metric (in spherical):

$$g_{\mu\nu} = \begin{pmatrix} 1 - \frac{2GM}{rc^2} & & & \\ & -\left(1 - \frac{2GM}{rc^2}\right)^{-1} & & \\ & & -r^2 & \\ & & & -r^2 \sin^2 \theta \end{pmatrix} \quad (14.1)$$

where empty places are zeroes. We'll define the time t to be measured at infinity, therefore the interval is

$$ds^2 = dx^\mu dx^\nu g_{\mu\nu} \quad (14.2)$$

$$= \left(1 - \frac{2GM}{rc^2}\right) (cdt)^2 - \left[\left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 + (rd\theta)^2 + (r \sin \theta d\phi)^2 \right] \quad (14.3)$$

If we define $\frac{2GM}{rc^2} = R_s$ then

$$ds^2 = \left(r - \frac{R_s}{r}\right) (cdt)^2 - \left[\left(1 - \frac{R_s}{r}\right)^{-1} dr^2 + \dots \right] \quad (14.4)$$

and we can use this to define the proper time

$$\tau = \frac{ds}{c} \quad (14.5)$$

Let's look at the following scenario: we have a mass M at position $r = 0$, and our friend Yossi is located at position $\mathbf{r}$. We're located at infinity. we measure a time interval dt . What time interval τ did Yossi measure? We look at the interval between the two events. both events happened at the same position so $dr = d\theta = d\phi = 0$. This means that

$$ds^2 = \left(1 - \frac{R_s}{r}\right) (cdt)^2 \quad (14.6)$$

therefore

$$\tau = \frac{ds}{c} = \sqrt{1 - \frac{R_s}{r}} dt \quad (14.7)$$

if we define $\Delta r = r - R_s$ then

$$d\tau = dt \sqrt{\frac{\Delta r}{r}} \quad (14.8)$$

and if we look at some numerical examples

$\frac{\Delta r}{r}$	$\frac{\Delta \tau}{\tau}$
0.5	1/1.4
0.1	$\sim 1/3$
0.01	1/10
10^{-4}	1/100

Let's give Yossi a flashlight and throw him into a black hole. We could have thrown the flashlight on its own, but that would be less fun. What we'll see is that Yossi's radius asymptotically approaches R_s but never crosses it. If we look at what the flashlight does, we'll notice the wavelength of the light

increases and the magnitude of the light decreases. If we define the period of the wavelength of the light to be T_0 and the frequency to be ν_0 then the period at infinity is

$$T_\infty = \frac{1}{\sqrt{1 - \frac{R_s}{r}}} T_0 \quad (14.9)$$

and

$$\nu_\infty = \nu_0 \sqrt{1 - \frac{R_s}{r}} \quad (14.10)$$

and the wavelength would be

$$\lambda_\infty = \frac{\lambda_0}{\sqrt{1 - \frac{R_s}{r}}} \quad (14.11)$$

if we look at a source of light which sends pulses of photons with frequency ν_0 at the rate $\kappa(s^{-1})$ then the power output of it is

$$P_0 = h\nu_0\kappa \quad (14.12)$$

and at infinity

$$P_\infty = P_0 \left(1 - \frac{R_s}{r}\right) \quad (14.13)$$

Let's now look at the behavior of light near black holes. To do that we'll look at the case of the interval being zero, $ds = 0$:

$$0 = \left(1 - \frac{R_s}{r}\right) (cdt)^2 - \left(1 - \frac{R_s}{r}\right)^{-1} dr^2 + dots \quad (14.14)$$

for a radial path $d\theta = d\phi = 0$ and we'll find

$$\frac{dr}{dt} = \pm \left(1 - \frac{R_s}{r}\right) c \quad (14.15)$$

but does this not violate the principle that light moves at the speed of light? No. This is the speed we observe as an external observer at infinity. the velocity as experienced by the light itself is

$$v = \frac{dl}{d\tau} = \frac{dr}{dt} \frac{\frac{1}{\sqrt{1 - \frac{R_s}{r}}}}{\frac{1}{\sqrt{1 - \frac{R_s}{r}}}} = \frac{\frac{dr}{dt}}{1 - \frac{R_s}{r}} = c \quad (14.16)$$

5.2 Thermodynamics of Black Holes

¹ The 2nd law of thermodynamics states that entropy increases, i.e $ds \geq 0$. The entropy of a black hole is proportional to its surface area $S_{BH} \propto A_{BH}$. If mass is absorbed by a black hole, its mass increases. This in turn increases the radius which increases the surface area, and thus the entropy increases. A more accurate calculation done by Hawking and Bekenstein yields

$$S_{BH} = k_B \frac{A}{4l_p^2} \quad (14.17)$$

where

$$l_p = \sqrt{\frac{G\hbar}{c^3}} \quad (14.18)$$

for a Schwarzschild black hole, $A = 4\pi R_s^2$. The black hole having an entropy which changes when mass is absorbed implies that it might have a definable temperature, which itself implies that we may

¹This is a very complex subject so we won't go deep into it but it's very interesting so we'll briefly cover it. Some interesting subtopics we won't cover are the information paradox, the Unruh effect, and Hologram Universe.

be able to measure a radiation being emitted from the black hole as a black body. We can define temperature using statistical mechanics to be

$$T = \frac{dU}{dS} = \frac{dU}{dM} \left(\frac{dS}{dM} \right)^{-1} \quad (14.19)$$

as we are looking at a Schwartzchild black hole we can define $U = Mc^2$ and find

$$S = k_B \frac{A}{l_p^2} = k_B \frac{4\pi R_s^2}{4l_p^2} = k_B \frac{\pi}{l_p^2} \left(\frac{2GM}{c^2} \right)^2 \quad (14.20)$$

therefore

$$\frac{dS}{dM} = k_B \frac{8\pi GM}{\hbar c} \quad (14.21)$$

and

$$\frac{dU}{dM} = c^2 \quad (14.22)$$

inserting back into 14.19

$$k_B T = \frac{\hbar c^3}{8\pi GM} \quad (14.23)$$

if we plug in numbers then the temperature is

$$T = 60 \text{ nK} \left(\frac{M}{M_\odot} \right)^{-1} \quad (14.24)$$

now since the black hole has a temperature, it'll radiate as a black body. How long would it take for a black hole then to radiate away all its mass?

$$\frac{dE}{dt} = \frac{dMc^2}{dt} = c^2 \dot{M} \quad (14.25)$$

but on the other hand

$$\frac{dE}{dt} = 4\pi R_s^2 \sigma T^4 \propto M^2 \cdot \left(\frac{1}{M} \right)^4 \propto \frac{1}{M^2} \quad (14.26)$$

which gives us a differential equation whose solution is

$$t_{\text{evap}} = \frac{5120\pi G^2 M^3}{\hbar c^4} \quad (14.27)$$

inserting the numbers find

$$t_{\text{evap}} = 2 \times 10^{67} \text{ yr} \left(\frac{M}{M_\odot} \right)^3 \quad (14.28)$$

so black holes practically never evaporate. For a black hole to evaporate within the age of the universe, they'd have to have mass

$$M \leq 10^8 \text{ g} \quad (14.29)$$

* * *